

Calculated Fields, Parameters & LOD Expressions

Q1. What is a Calculated Field and why is it important in business dashboards?

Ans: A **Calculated Field** in Tableau is a field that is created using a formula to perform calculations on existing data. It allows users to generate new values from existing dimensions and measures without modifying the original data source.

It is important because:-

Creates Custom Metrics

- Businesses can create KPIs such as Profit Margin, Growth Rate, Average Revenue per Customer, etc.

Improves Data Analysis

- Helps uncover trends and patterns that are not directly available in the raw data.

Supports Better Decision-Making

- Managers can monitor meaningful business indicators instead of relying only on basic data fields.

Reduces Manual Calculations

- Calculations are automated within the dashboard, saving time and reducing errors.

Enhances Dashboard Insights

- Enables dynamic analysis, comparisons, and performance tracking across departments, products, or regions.

Q2. Why can't parameters change visuals directly in Tableau?

Ans: Parameters cannot change visuals directly in Tableau because they only store a single user-defined value and do not automatically affect the data or chart. A parameter must be connected to a calculated field, filter, reference line, or other Tableau feature to influence the visualization.

Parameters can't change visuals directly in Tableau because:-

- A **parameter** acts like an input box that holds a value.
- By itself, it has no relationship with the data in the view.
- Tableau needs a **calculated field or logic** that references the parameter value to determine how the visualization should change.

Q3. When should EXCLUDE LOD be used?

Ans: **EXCLUDE LOD (Level of Detail)** is used when you want to **remove one or more dimensions from the current view's level of detail and perform calculations at a higher level of aggregation.**

EXCLUDE LOD should be used when:-

Compare a detailed value with a higher-level total

- Example: Compare each product's sales with the total sales of its category.

Calculate averages or totals without a specific dimension

- Example: Show each customer's sales while also displaying the average sales across all customers in the same region.

Analyze contribution to a group

- Example: Determine how much each product contributes to overall category sales.

Q4. Difference between FIXED and INCLUDE LOD expressions?

Ans:

FIXED LOD

Calculates values at a specified level of detail,

INCLUDE LOD

Calculates values by adding specified dimensions to the

FIXED LOD

regardless of the dimensions present in the view.

Syntax: { FIXED
[Dimension] :
AGG([Measure]) }

Independent of the visualization's dimensions (except context filters).

Used when a calculation must always be performed at a fixed level.

Example: Total sales per region, regardless of product or customer displayed.

Example

FIXED LOD

{ FIXED [Region] : SUM([Sales]) }

Returns the total sales for each region, even if the view contains Product or Customer dimensions.

INCLUDE LOD

level of detail already present in the view.

Syntax: { INCLUDE
[Dimension] :
AGG([Measure]) }

Depends on the dimensions currently in the view.

Used when a more detailed calculation is needed than the current view provides.

Example: Calculate customer-level sales and then aggregate them at the region level.

INCLUDE LOD

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{ INCLUDE [Customer Name] : SUM([Sales]) }
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Adds Customer Name to the level of detail, allowing calculations at the customer level before aggregating results in the view.